

VDF Assignment-1

Objective

The objective of this assignment is to gain hands-on experience with UNIX and TCL.

You can go through the following tutorials to get a head start on UNIX and TCL:

UNIX:

https://www.google.com/url?q=https://www.youtube.com/watch?v%3DztPFMRfpPfk&sa=D&source=editors&ust=1724175316641209&usg=AOvVaw1H8lQA2q1MW999rL_VGOnl

TCL:

https://www.google.com/url?q=https://www.youtube.com/watch?v%3D1fPNZstiL4o&sa=D&source=editors&ust=1724175316641507&usg=AOvVaw0MdImLBNTi10UF98-kp_s6

Instructions

- These assignments need to be done individually
- Read the instructions given in the questions carefully
- Each student will submit a ZIP file containing TCL (.tcl) and Shell (.sh) files.
- Write comments in the code and also explain them wherever applicable.
- Do not copy code/explanation from others or the internet, not only in terms of content but also in terms of presentation.
- **Copying/Plagiarism is strictly prohibited. Institute copying/plagiarism policy will apply with no exception.**

Questions

1. Write a TCL script to analyze a text file and print the following output.
 - Number of words.
 - Number of lines.
 - Number of characters (excluding spaces).
 - The longest word in the file.
 - The shortest word in the file.

The code should also print your name and your roll number as shown in the sample output below (**without this information, you will receive a zero**).

Note: Do not use any Unix/shell commands inside your TCL code

Your code should be run on the UNIX platform, as shown in the following screenshot.

Your code file name should be: code.tcl

The text file is: sample.txt (uploaded on Google Classroom)

```
yashw@Phoenix:~/VDF_assign$ tclsh code.tcl sample.txt
Total Words:      62
Total Lines:      5
Total Characters: 337

Longest Word:  Independence
Shortest Word: a

Prepared by: Yashwardhan Tyagi
Roll Number: Phd24109
yashw@Phoenix:~/VDF_assign$
```

The format of the output should be as shown above to get marks.

2. Write a Shell script to analyze a given text file and print the following output

- Number of words.
- Number of lines.
- Number of characters (excluding spaces).
- The longest word in the file
- The shortest word in the file

The code should also print your name and your roll number as shown in the sample output below (without this information, you will receive a zero).

Note: Do not use any TCL commands inside your shell script.

Your code should be run on the UNIX platform, as shown in the following screenshot.

Your code file name should be: code.sh

The text file is: sample.txt (uploaded on Google Classroom)

```
garima@Garry:~/assign_1$ bash code.sh sample.txt
File Name      : sample.txt
Total Lines    : 5
Total Words    : 62
Total Characters : 337
Longest Word   : Independence
Shortest Word  : a
Name          : Garima
Roll Number: PhD25102
```

The format of the output should be as shown above to get marks.

You will not be evaluated based on the sample.txt file provided in Google Classroom. There will be a separate text file for the evaluation. The sample test file is for your reference.